Lithium Prospectivity Mapping in Québec

Identifying priority exploration zones from SIGÉOM data

An exploratory analysis using Python, QGIS, and SIGÉOM open databases

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Year: 2026
Tools used: Python — QGIS — SIGÉOM

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1. Introduction

The energy transition has placed lithium at the center of North American and European industrial strategies. Used in the manufacturing of batteries for electric vehicles and for stationary electricity storage, this metal is the subject of growing demand that is pushing several mining jurisdictions to reassess their potential. Québec is one of the North American territories where lithium exploration has accelerated in recent years, particularly in the James Bay region and in certain sectors of Abitibi, two areas associated with granitic pegmatite geological formations that are generally favorable to this type of mineralization.

Québec has a particular advantage for studying this dynamic: the SIGÉOM database, managed by the Ministère des Ressources naturelles et des Forêts, provides the public with a very detailed set of information on mining projects, geological indices, and exploration targets recorded across the territory. This openness of data makes it possible to build, using relatively simple tools, an initial spatial reading of Québec's lithium activity, without resorting to field geological expertise.

This work follows that logic. It does not claim to produce a professional geological assessment of mining potential, but instead proposes an exploratory analysis combining data processing, mapping, and a reading informed by economic geography. The objective is to show how SIGÉOM databases can be cross-referenced with one another in order to locate existing lithium projects, identify lithium-associated mineralized occurrences beyond already-known projects, rank these occurrences using a prospectivity score, and then relate them to the exploration targets recorded in the same database.

The research question structuring this analysis is the following: to what extent does cross-referencing lithium projects, GMLI occurrences, and SIGÉOM exploration targets make it possible to identify priority zones for future lithium prospecting in Québec?

To answer this question, the report first presents the data and method used, then successively details the identification of lithium projects, the analysis of GMLI occurrences, the construction of the prospectivity score, the cross-referencing with exploration targets, the identification of priority prospecting zones, and the final cartographic analysis. The limitations of the approach are then discussed before the conclusion.

2. Data, Tools and General Method

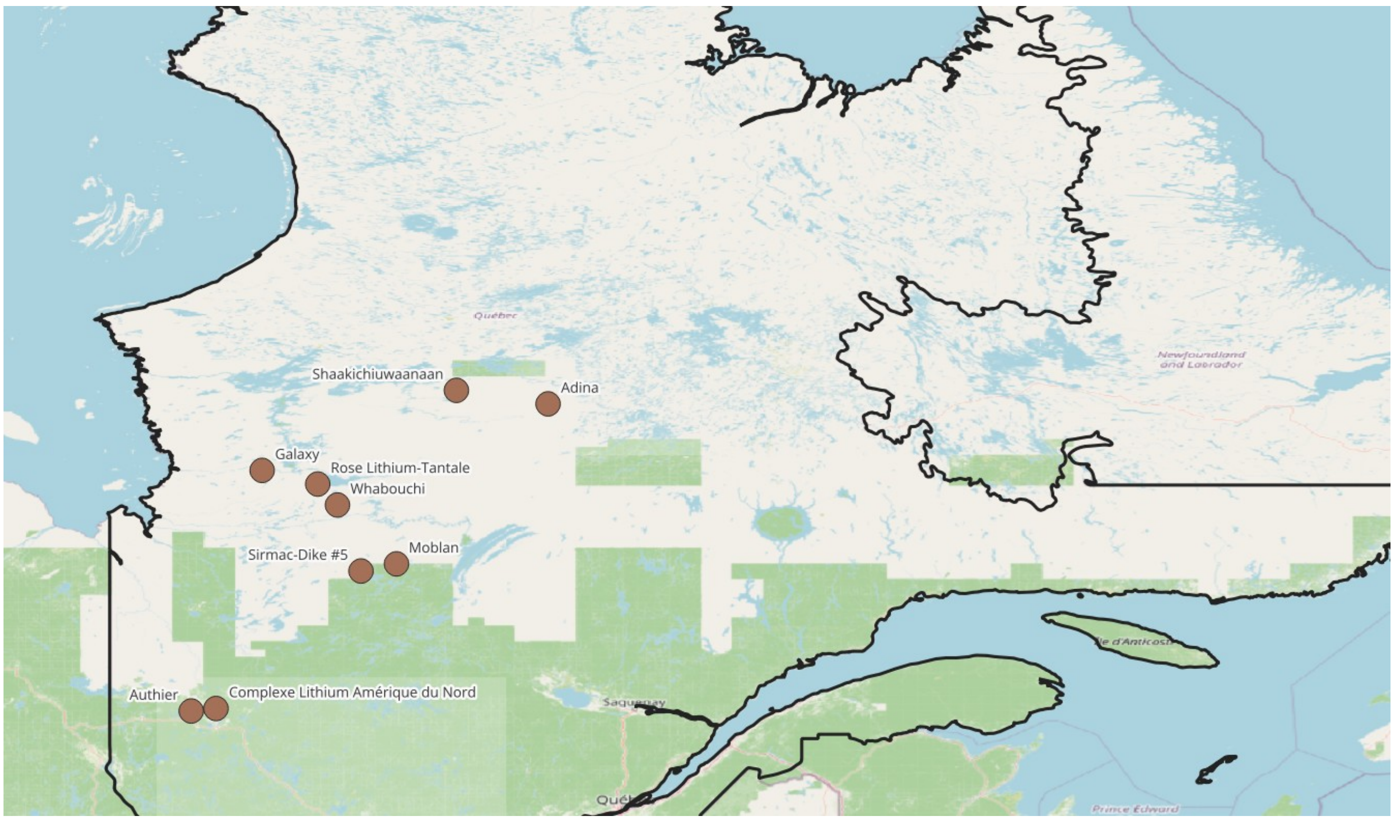
The analysis relies on three public SIGÉOM databases: the “Mines and Projects” database, which records mining projects at various stages of progress; the “Metallic Substances” database, which groups together mineralized occurrences identified across the territory, independently of their project status; and the “Exploration Targets” database, which brings together sectors that have been flagged by geologists, without this flagging necessarily constituting confirmation of an exploitable resource.

Data processing was carried out using Python, which was used to filter substance columns, extract relevant occurrences, build the prospectivity score, calculate distances between points, and export the summary files used in this report. The mapping was produced in QGIS, based on an OpenStreetMap basemap and, for the general location map, a Natural Earth layer used to overlay the administrative boundary of Québec.

It is important to clarify from the outset the status of this approach. None of the steps described below constitutes geological proof of the existence of an exploitable deposit. The work aims to organize and cross-reference already public data in order to surface zones of interest, that is, sectors where several favorable signals overlap spatially. These results should be read as a screening aid tool, capable of guiding further investigation, and not as a conclusion on the actual mining value of the identified sites.

3. Identification of Recorded Lithium Projects

The first step of the analysis consisted of querying the “Mines and Projects” database using the substance columns, searching for the Li2O code, which designates lithium oxide used as a marker for lithium content in geological reports. This filtering identified nine projects distributed across Québec territory: Complexe Lithium Amérique du Nord, Galaxy, Rose Lithium-Tantale, Moblan, Whabouchi, Authier, Sirmac-Dike #5, Shaakichiuwaanaan, and Adina.



Map 1 — General location of the 9 lithium projects recorded in SIGÉOM's “Mines and Projects” database.

The spatial distribution of these nine projects is not homogeneous. Seven of them are located within the Plan Nord boundary, namely Galaxy, Rose Lithium-Tantale, Moblan, Whabouchi, Sirmac-Dike #5, Shaakichiuwaanaan, and Adina, while only two projects lie outside this boundary: Complexe Lithium Amérique du Nord and Authier, both located in Abitibi.

This concentration within Plan Nord is not a statistical coincidence. Plan Nord constitutes a territorial framework that has structured a significant share of Québec's mining activity for more than a decade, by organizing access to infrastructure, energy development, and, more broadly, the conditions of accessibility to territories that are otherwise distant from major transport routes. The majority presence of lithium projects within this boundary illustrates a classic logic in economic geography: the location of mining activities does not depend solely on the nature of the resource sought, but also on the institutional and logistical environment that makes that resource accessible and exploitable. The territory functions here as a genuine factor of economic organization, not as a simple neutral backdrop over which projects would be distributed at random.

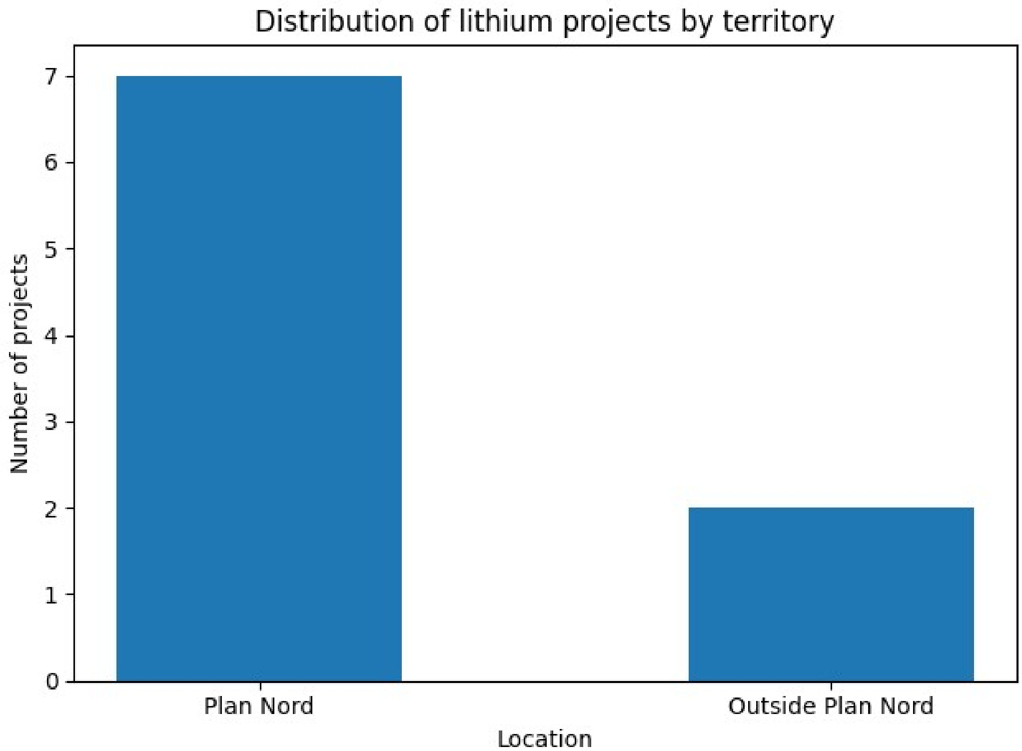


Chart 1 — Distribution of lithium projects according to whether they fall within Plan Nord.

Examination of the promoters associated with these nine projects confirms a certain diversity of actors, although one promoter, Elevra Lithium ltd., holds three of the nine recorded projects. The other six projects each belong to a distinct promoter: Rio Tinto Lithium Canada inc., Corporation Lithium Éléments Critiques, Nemaska Lithium inc., Vision Lithium inc., Ressources PMET inc., and Li-FT Power Ltd.

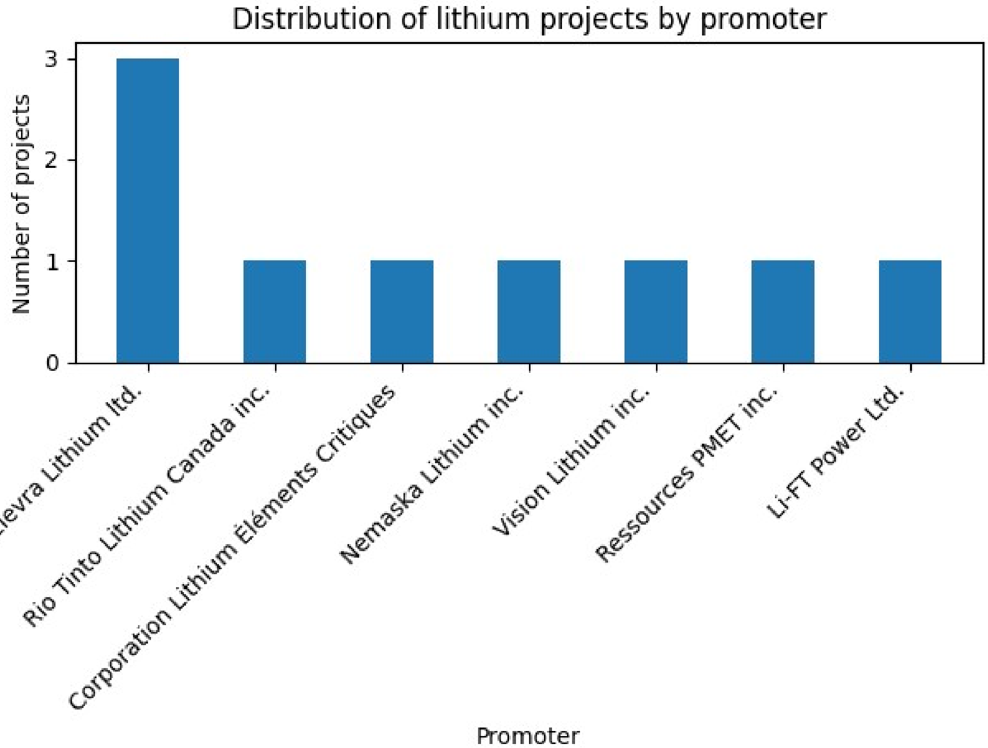


Chart 2 — Distribution of lithium projects by promoter.

This relatively concentrated structure around a single promoter resembles a form of polarization of actor strategies around certain mining spaces already identified as promising. An actor holding several projects in the same region tends, in effect, to reinforce the territorial specialization of that zone around the lithium sector, by concentrating investment, infrastructure, and accumulated expertise there. The SIGÉOM database also contains a status field for each project, but the exact meaning of these codes could not be fully documented at this stage; any reading of the distribution by status must therefore be handled with caution and is not developed further in this report.

4. Analysis of GMLI Occurrences

The nine projects identified previously represent only a fraction of the information available on lithium in Québec. To broaden the analysis, SIGÉOM's “Metallic Substances” database was used. This database records a much larger set of mineralized bodies, in the form of 9,291 rows spread across 93 columns, which are not limited to projects under development but also include indices and showings at very different stages of progress.

Python-based filtering was used to extract, based on the GMLI code, the full set of occurrences associated with lithium. This extraction produced 159 occurrences, exported to the file indices_lithium_gmli_quebec.csv. It is important to distinguish these occurrences from the projects recorded in the previous chapter: a GMLI occurrence may correspond to a simple indication with no documented work, to an indication that has been the subject of exploration work, or to a showing for which a resource has been partially estimated. The summary provided indicates that, of the 159 occurrences, 85 correspond to an undeveloped indication, 61 to a worked indication, 12 to a showing, and only one to an active mine.

This distribution confirms that the vast majority of GMLI occurrences remain at an exploratory or weakly documented stage. Their interest for this report lies precisely in that: they make it possible to broaden the spatial analysis beyond sites already engaged in a project, and to observe whether the zones where these occurrences concentrate overlap with sectors already identified by industry.

5. Building the Prospectivity Score

To rank the 159 GMLI occurrences according to their potential interest, a prospectivity score was built from several cumulative criteria: the presence of the GMLI code itself, the fact that lithium constitutes the occurrence's main substance, membership in a pegmatite-type typology, the existence of an estimated resource, the status of the mineralized body (showing or worked indication), the presence of substances generally associated with lithium such as tantalum, cesium, beryllium, niobium, or rubidium, and finally geographic proximity to an already-known lithium project.

The nature of this indicator must be emphasized: it is in no way an official measure nor geological proof. The score serves solely to rank occurrences relative to one another, based on signals observable in the database, in order to direct attention toward the sectors that combine the greatest number of favorable characteristics.

Potential class	Number of occurrences
Strong potential	102
Moderate potential	55
Low potential	2
Total	159

Table 1 — Distribution of GMLI occurrences by potential class.

The result shows a clear majority of occurrences classified as strong potential, 102 out of 159, compared with 55 as moderate potential and only 2 as low potential. This distribution is largely explained by the very nature of the sample: the occurrences were pre-selected on the GMLI criterion, which means they already share several of the characteristics sought by the score. The typology associated with granitic pegmatites dominates very widely, with 148 occurrences out of 159, corresponding to the most common type of lithium mineralization in the Québec geological context.

Examination of the ranking shows that the ten occurrences obtaining the maximum score are located, in almost every case, in the immediate vicinity of an already-recorded lithium project: Cyr-Lithium near Galaxy, Lac Moblan Ouest near Moblan, Rose near Rose Lithium-Tantale, Authier near Authier, Sirmac near Sirmac-Dike #5, Whabouchi near Whabouchi, and CV5, CV13, and Adina Main respectively near Shaakichiuwaanaan and Adina. This result is not surprising in itself, since proximity to a known lithium project is one of the criteria entering into the score calculation. This correspondence nevertheless constitutes an internal consistency indicator for the method: the best-ranked occurrences are often located in sectors already structured around lithium projects. This does not scientifically validate the score, but shows that it captures signals already partially documented in the SIGÉOM data.

6. Crossing with Exploration Targets

The third database used is SIGÉOM's "Exploration Targets" database, which groups together 5,622 targets described across 105 columns. An exploration target corresponds to a sector that has been flagged by geologists as presenting an interest for further work, without this flagging being equivalent to confirmation of a deposit.

An initial attempt to filter by the keyword "Li" proved unsuitable: this string of letters appears in many terms unrelated to lithium, for example "appliquée" or "population," which generated a high number of false positives. The filtering was therefore reformulated based on a list of more specific keywords: lithium, pegmatite, spodumene, tantalum, cesium, rubidium, and niobium. This methodological correction made it possible to retain 44 exploration targets actually related to lithium or pegmatites, exported to the file cibles_exploration_lithium_pegmatite.csv. These 44 targets obviously do not constitute proof of a deposit; they are merely compatible, through their descriptive content, with an exploratory lithium prospecting approach.

A spatial cross-referencing was then carried out between the 159 scored GMLI occurrences and the 44 targets thus retained. For each occurrence, the nearest target was identified and the corresponding distance was calculated in kilometers, then sorted into four proximity classes.

Proximity class	Number of occurrences
Very close to a target	41
Close to a target	12
Within a 100 km radius	38
Far from targets	68
Total	159

Table 2 — Proximity of GMLI occurrences to exploration targets.

In total, 91 occurrences out of 159, or a little over half of the sample, are located less than 100 km from an exploration target related to lithium or pegmatites, adding together the "very close," "close," and "within a 100 km radius" classes. This proportion suggests that the occurrences identified by the score are not randomly scattered across the territory, but largely fall within sectors already flagged by previous exploration work.

The combined ranking of score and distance to targets highlights a group of occurrences of particular interest for an exploratory reading. Adina Main, already classified as strong potential, is located about 1 km from the "Adina" target; CV13 and CV5 are located respectively about 12, 15, 16, and 17 km from target 10-SB-6326; Cyr-Lithium is located a little over 38 km from the "Béryl" target. Other occurrences classified as strong potential by the score, such as Whabouchi, Authier, or Lac Moblan Ouest, are on the other hand relatively far from any recorded target, at distances exceeding 100, or even 200 km. This contrast illustrates an important limitation of the approach: a high score based on the intrinsic characteristics of an occurrence is not necessarily accompanied by proximity to a documented exploration target, which may stem as much from the actual history of local exploration as from gaps in the coverage of the targets database. The full cross-ranking list appears in the annex.

7. Identifying Priority Prospecting Zones

The previous results make it possible to move from a point-by-point reading of occurrences to a territorial reading of prospectivity. The objective is no longer only to identify the best-ranked lithium occurrences, but to locate sectors where several favorable signals overlap: a high prospectivity score, proximity to an exploration target, proximity to a known lithium project, and spatial concentration of occurrences.

In this logic, occurrences are not interpreted in isolation. They are grouped into zones of interest, in order to identify sectors that may justify further prospecting work. These zones do not constitute confirmed deposits. They represent spaces favorable to future exploration, built from open data and a spatial cross-referencing method.

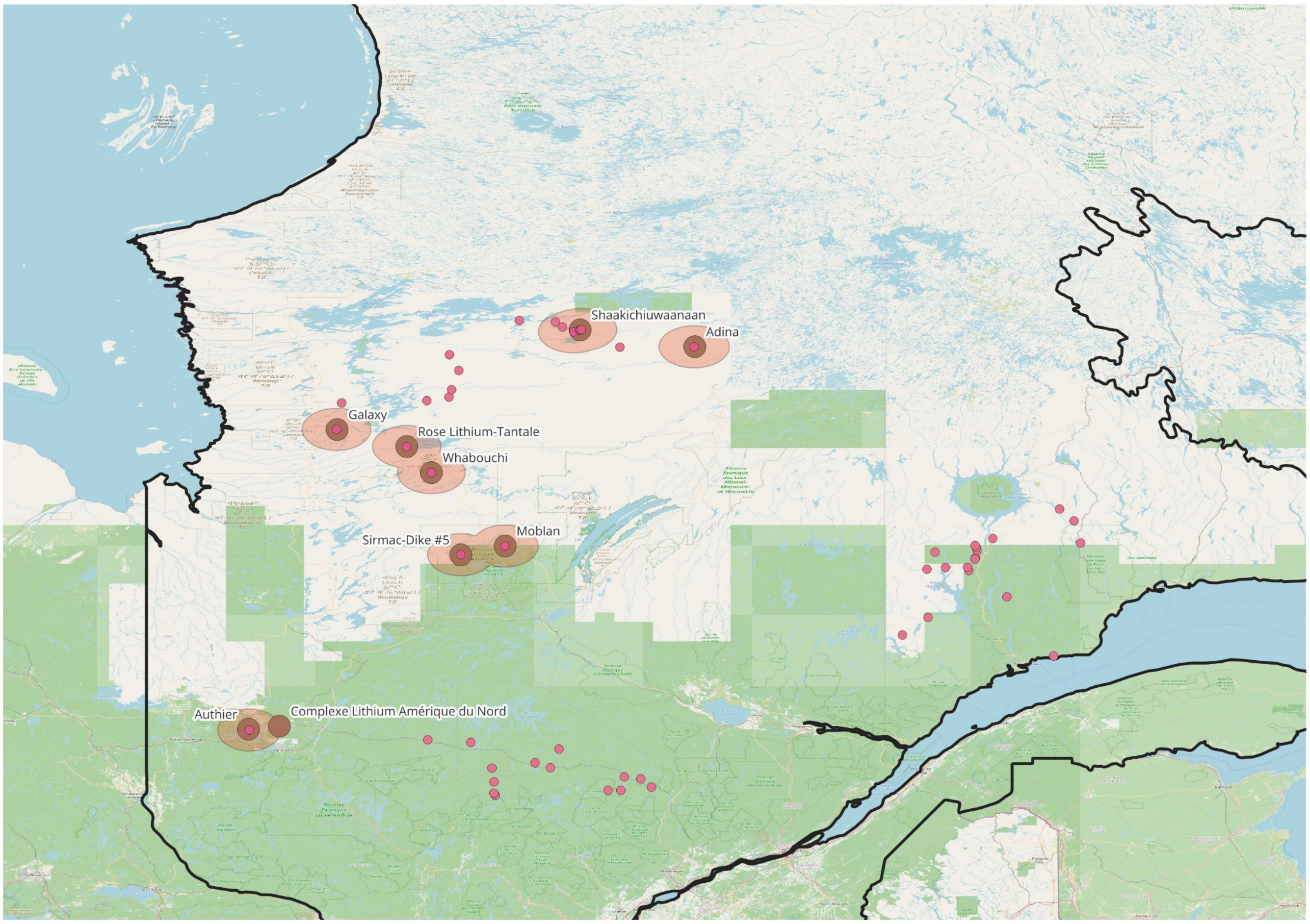
Rank	Priority zone	Favorable signals	Level of interest	Prospective reading
1	Adina / Adina Main	Maximum score, very close target, known project	Very high	Priority sector to investigate further
2	CV5 / CV13 / Shaakichiuwaanaan	Maximum score, concentration of occurrences and nearby targets	Very high	Zone of strong exploratory concentration
3	Galaxy / Cyr-Lithium	Maximum score, existing project, structured lithium context	High	Sector consistent with an existing mining dynamic
4	Rose / Whabouchi	High score, known projects, regional proximity of targets	High	Mining hub worth consolidating
5	Moblan / Sirmac-Dike #5	High score, proximity to known projects	Moderate to high	Interesting zone but requiring further documentation
6	Authier / Complexe Lithium Amérique du Nord	High score, sector outside Plan Nord	Moderate to high	Strategic interest for comparing northern and Abitibi dynamics

Table 3 — Identified priority lithium prospecting zones.

The identification of priority zones makes it possible to move beyond a point-by-point reading of occurrences. The Adina / Adina Main and CV5 / CV13 sectors stand out as the most robust, since they combine a maximum prospectivity score with strong proximity to exploration targets. They are therefore the zones most directly compatible with a priority prospecting logic.

The Galaxy / Cyr-Lithium, Rose / Whabouchi, and Moblan / Sirmac-Dike #5 sectors also appear important, as they fall within mining hubs already structured around known lithium projects. Finally, the Authier / Complexe Lithium Amérique du Nord sector is of particular interest, as it allows a comparison between the Plan Nord dynamic and that of Abitibi.

This ranking should not be understood as confirmation of a deposit. It instead constitutes a decision-support reading, intended to highlight the sectors where the overlap of signals justifies further investigation.



Map 3 — Priority lithium prospecting zones in Québec.

The map highlights the main priority zones identified based on the prospectivity score, proximity to exploration targets, and the presence of known lithium projects. The circles represent 30 km zones around the best-ranked GMLI occurrences. This representation does not confirm the existence of deposits, but makes it possible to identify sectors where several favorable signals overlap.

This reading makes it possible to rank sectors of interest, but it must be interpreted with caution, given the limitations inherent to open data and the absence of field geological validation.

Source: SIGÉOM, personal Python/QGIS processing, 2026.

8. Limitations

Several limitations must be noted before concluding. The prospectivity score relies on criteria chosen a priori, whose weighting remains relatively simple and does not necessarily reflect the relative importance that specialized geologists would assign to each of them. The keyword filtering, although improved after correcting the “Li” false positive, remains sensitive to the exact wording of the descriptions contained in the database, which may lead to omitting certain relevant targets or retaining others only marginally.

The distance calculation used for cross-referencing occurrences and targets relies on an approximate geographic distance between coordinates. It does not account for terrain relief, actual ground accessibility, roads, or logistical constraints. At the cartographic scale used for this report, the entire Québec territory is represented in a relatively compact way, which can give an impression of proximity between points that are in reality separated by several dozen kilometers of poorly accessible territory.

Finally, and this is the most fundamental limitation, this work does not rely on any drilling data, any detailed geochemical analysis, or any study of bedrock geology. Only field geological expertise, drawing on data far more detailed than what is available in SIGÉOM in tabular form, would make it possible to truly conclude that exploitable mining potential exists in the zones of interest identified here.

The proposed ranking relies on a simplified multi-criteria approach. It allows zones to be compared with one another, but does not replace a detailed structural analysis, geochemical surveys, drilling, or field validation.

9. Conclusion

This report has proposed an exploratory reading of lithium in Québec based on public SIGÉOM data. The analysis made it possible to identify nine lithium projects, seven of which are located within Plan Nord, then to broaden the scope to 159 GMLI occurrences distributed across three potential classes built from a prospectivity score. Cross-referencing these occurrences with 44 exploration targets retained after correcting the keyword filtering showed that more than half of them, 91 out of 159, are located less than 100 km from a target already flagged by geologists.

The main contribution of this study lies in the shift from descriptive mapping to an exploratory targeting approach. By combining known projects, GMLI occurrences, exploration targets, and the prospectivity score, the analysis makes it possible to identify several priority prospecting zones, notably around Adina, CV5/CV13, Galaxy/Rose/Whabouchi, Moblan/Sirmac-Dike #5, and Authier. These sectors do not constitute confirmed deposits, but zones where several favorable signals overlap and which warrant particular attention from the perspective of future exploration.

This spatial organization illustrates, at the scale of Québec, the classic mechanisms of economic geography: the location of mining activities depends not only on the presence of the resource, but also on the accessibility of the territory, the history of exploration, and the strategies of promoters already engaged in the region.

The interest of this approach lies above all in its method: it shows that it is possible, using open data and relatively accessible tools, to build a coherent spatial reading of a contemporary mining issue, without specialized geological expertise. The results obtained remain favorable signals rather than proof, and naturally call for further work: integration of drilling and detailed geochemical data, consideration of bedrock geology, or analysis of the actual accessibility of the identified sectors based on their distance to existing roads and infrastructure networks.

Data Sources

All of the data used in this report comes from SIGÉOM, the public geomining database managed by Québec's Ministère des Ressources naturelles et des Forêts, via the Géologie Québec portal. Three databases from this platform were used: “Mines and Projects,” “Metallic Substances,” and “Exploration Targets.”

- SIGÉOM / Géologie Québec — Québec's public geomining data, “Mines and Projects,” “Metallic Substances,” and “Exploration Targets” databases : https://sigeom.mines.gouv.qc.ca/
- Géologie Québec : https://gq.mines.gouv.qc.ca/
- Natural Earth — “Admin 1 — States, Provinces” administrative layer, used for the outline of Québec : https://www.naturalearthdata.com/
- OpenStreetMap — basemap used in QGIS : https://www.openstreetmap.org/

Data extraction, filtering, construction of the prospectivity score, and distance calculations were carried out through personal processing in Python. The mapping was produced in QGIS, based on the layers mentioned above. See GitHub.

Annex 1 — Cross-Ranking of the Ten Highest-Scoring Occurrences

The table below presents, for the ten occurrences obtaining the maximum prospectivity score, the nearest exploration target, the corresponding distance, and the associated proximity class. Some occurrence names appear more than once, as they may correspond to distinct mineralized bodies or distinct records in the SIGÉOM database.

Occurrence	Score	Nearest target	Distance (km)	Proximity class
Adina Main (Jamar)	14	(6) Adina (14-AB-6048A)	1.01	Very close to a target
CV13	14	10-SB-6326	11.85	Very close to a target
CV5	14	10-SB-6326	15.09	Very close to a target
CV5	14	10-SB-6326	15.60	Very close to a target
CV5	14	10-SB-6326	16.55	Very close to a target
Cyr-Lithium (James Bay Lithium)	14	Béryl	38.11	Close to a target
Rose	14	FS-3292	68.27	Within a 100 km radius
Whabouchi	14	FS-3292	103.33	Far from targets
Authier	14	Indice Kekek (15-AM-200B)	170.54	Far from targets
Lac Moblan Ouest	14	PR-6073	219.24	Far from targets

Annex 2 — Strategic Pre-Analysis of Priority Zones

Based on the prospectivity score, proximity to known lithium projects, and distance to exploration targets, a strategic pre-analysis was built in order to rank priority zones according to their potential interest for an investor. This reading does not constitute a financial valuation, but a first screening grid intended to distinguish the most robust sectors from those requiring further verification.

Concretely, each priority zone was assessed using three signals. The first is the prospectivity score, which summarizes the favorable geological characteristics already identified in the SIGÉOM data. The second is proximity to a known lithium project, which indicates whether the zone falls within an already-structured mining sector. The third is the distance to the nearest exploration target, which makes it possible to assess whether the occurrence is also associated with a sector already flagged for future work.

The logic applied is simple: a zone combining a high score, strong proximity to a lithium project, and a nearby exploration target is considered more robust. Conversely, a zone ranked very well geologically but far from exploration targets is kept in the ranking, but with a more cautious level of investor interest.

This grid is deliberately indicative. It makes it possible to rank zones according to their apparent strategic interest, but it does not replace a complete financial analysis incorporating resources, grades, operating costs, infrastructure, lithium prices, or regulatory risks.

Priority zone	Associated project	Min. distance to project	Min. distance to a target	Spatial signal	Exploration risk	Investor interest
Adina / Adina Main	Adina	0.55 km	1.01 km	Maximum score + very close target	Moderate	High
CV5 / CV13 / Shaakichiuwaanaan	Shaakichiuwaanaan	0.00 km	11.85 km	Maximum score + concentration of occurrences	Moderate	High
Galaxy / Cyr-Lithium	Galaxy	0.62 km	38.11 km	Nearby project + nearby target	Moderate to high	Moderate to high
Rose / Whabouchi	Rose / Whabouchi	0.00 km	68.27 km	Structured lithium hub	Moderate to high	Moderate to high
Moblan / Sirmac-Dike #5	Moblan / Sirmac-Dike #5	0.32 km	219.24 km	Nearby project but distant target	High	Moderate
Authier / Complexe Lithium Amérique du Nord	Authier / CLAN	1.39 km	170.54 km	Nearby project but distant target	High	Moderate